

📖 CHEMISTRY CH.9 — IDEAL GAS LAW — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

📖 Topics in this chapter:

1. Combustion & Balanced Equations
2. Temperature Scales & Conversions
3. Gases: Properties & Identification
4. Gas Laws & Ideal Gas Equation
5. States of Matter & Phase Changes
6. Avogadro's Number & Molecular Concepts
7. Molar Volume at NTP & Numerical Tricks
8. ⚠️ Traps / Key Differences Summary

1. Combustion & Balanced Equations

- Methane combustion: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$. 1 methane molecule se 1 CO_2 aur 2 H_2O molecules bante hain. [Q1335]
- Ye complete combustion ka example hai — $\text{hydrocarbon} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$. [Q1335]

2. Temperature Scales & Conversions

- Freezing point of water: $0^\circ\text{C} = 32^\circ\text{F} = 273.15\text{ K}$. [Q1336]
- Boiling point of water: 100°C . [Q1336]
- Conversion formulas:
 - $\text{K} = ^\circ\text{C} + 273.15$
 - $^\circ\text{C} = (^\circ\text{F} - 32) \times 5/9$ [Q1336]
- Kelvin scale ka naam Lord Kelvin (William Thomson) ke naam pe rakha gaya hai. [Q1336]
- Maximum density of water: $4^\circ\text{C} = 277\text{ K}$ pe hoti hai — is temperature pe do opposing effects balance mein hote hain. [Q1339]
- Environmental effect: Lakes aur ponds upar se pehle freeze hote hain kyunki 4°C pe water densest hota hai. [Q1339]

3. Gases: Properties & Identification

- Chlorine (Cl_2): Greenish-yellow gas, toxic, corrosive, irritating to eyes/respiratory system; second-lightest halogen. [Q1340]
- Carbon Monoxide (CO): Odourless, colorless, incomplete combustion se banta hai. [Q1340]
- Hydrogen Sulphide (H_2S): Colorless, poisonous, corrosive, flammable; rotten egg smell; chalcogen-hydride gas. [Q1340]
- Iodine: Heaviest stable halogen; mineral found in some foods. [Q1340]
- Noble Gases: Monoatomic hote hain — He, Ne, Ar, Kr, Xe. [Q1341]
- Diatomic: H, F etc (atomic pairs). Polyatomic: Se, Ozone (O_3). [Q1341]
- CO_2 in Soda Water: Carbonation process se dissolve hoti hai; Henry's Law se describe ki jaati hai; pH 3–4; Pepsi, Coke sab mein hoti hai. [Q1337]

- Freon: Refrigerant ke roop mein use hota hai — soda water mein nahi. [Q1337]
- Nitrogen (N₂): Atmosphere ka ~78% hissa; non-metal. [Q1337]

4. Gas Laws & Ideal Gas Equation

- Ideal Gas Equation: $PV = nRT$. [Q1342, Q1347]
 - P = Pressure
 - V = Volume
 - n = moles of gas
 - R = Universal gas constant = $8.31 \text{ JK}^{-1} \text{ mol}^{-1}$
 - T = Temperature (Kelvin) [Q1342, Q1347]
- Time gas ke behavior ko describe nahi karta — $PV=nRT$ mein nahi aata. [Q1342]
- Boyle's Law: Constant temperature pe, $V \propto 1/P$; $PV = \text{constant}$. [Q1344]
- Gas constant (R): Ideal gas ke properties (P, T, V, quantity) ke beech relationship quantify karta hai; low pressure pe valid. [Q1347]
- Temperature increase pe molecular motion distribution:
 - Maximum possible speed badhti hai.
 - Speed distribution curve broaden hota hai.
 - Higher speed pe molecules ki sankhya badhti hai.
 - Same temperature pe heavier molecules slower hote hain lighter ones se.
 - Curve ke neeche ka area same rehta hai. [Q1348]

5. States of Matter & Phase Changes

- Evaporation: Liquid → Gas; ALL temperatures pe hota hai — sirf boiling point pe nahi. [Q1345, Q1346]
 - Surface phenomenon hai.
 - Surface area badhane se rate badhti hai.
 - Temperature, wind velocity, humidity pe depend karta hai. [Q1345, Q1346]
- Condensation: Gas → Liquid. [Q1346]
- Solidification: Liquid → Solid. [Q1346]
- Liquefaction: Gas → Liquid. [Q1346]
- Sedimentation: Water + insoluble mixture mein heavier components neeche baith jaate hain. [Q1338]
- Decantation: Liquid ko solid aur immiscible liquids se alag karna. [Q1338]
- Filtration: Solid aur fluid ko mixture se alag karna. [Q1338]
- Evaporation = vapourization of liquid into gas; NOT same as boiling. [Q1338, Q1345]

6. Avogadro's Number & Molecular Concepts

- Avogadro's Number: $N_0 = 6.022 \times 10^{23}$ (units: electrons, ions, atoms, molecules per mole). [Q1343]
- Avogadro's Law: Same temperature aur pressure pe, equal volumes of all gases mein same number of molecules hote hain. [Q1343]
- Speed distribution molecule ke mass pe depend karta hai — same T pe heavier molecules slower. [Q1348]

7. Molar Volume at NTP & Numerical Tricks

- NTP (Normal Temperature & Pressure) pe 1 mole kisi bhi ideal gas ka volume 22.4 L hota hai. [Q1349]
- Avogadro's Law: Same T & P pe, equal volumes mein equal number of molecules hote hain — isliye mass/volume ratio constant rehta hai for a given gas. Formula: $W_1/V_1 = W_2/V_2$. [Q1349]
- Numerical example: 10 g gas X ka volume NTP pe 5.6 L hai — molecular weight nikaalna hai.
 - Logic: Agar 10 g $\rightarrow$ 5.6 L, toh 22.4 L (1 mole) ka kitna mass hoga?
 - Calculation: $W_2 = (10 \text{ g} \times 22.4 \text{ L}) / 5.6 \text{ L} = 224/5.6 = 40 \text{ g/mol}$.
 - Trick: $22.4/5.6 = 4$, toh $10 \times 4 = 40$ directly aa jaata hai — time bachao exam mein. [Q1349]
- NTP pe 22.4 L/mol yaad rakho; STP pe bhi approximately same hota hai lekin exact values alag ho sakti hain (STP = 0°C, 1 atm; NTP = 20°C, 1 atm ya 298 K). [Q1349]

8. ⚠ Traps / Key Differences Summary

- ⚠ Trap: Evaporation sirf boiling point pe nahi — sabhi temperatures pe hota hai. [Q1345]
- ⚠ Trap: Temperature badhane pe "most probable motion wale molecules ka fraction increases" — ye statement galat hai. Actually curve broaden hone pe ye fraction ghat-ta hai. [Q1348]
- ⚠ Trap: Time gas behavior describe nahi karta — $PV=nRT$ mein nahi aata. [Q1342]
- ⚠ Trap: Water ka maximum density 4°C (277 K) pe hota hai — 0°C pe nahi. [Q1339]
- ⚠ Trap: Freezing point 273.15 K hai — 273 K exact nahi, 273.15 K exact value hai. [Q1336]
- ⚠ Trap: Freon = refrigerant; soda water mein CO₂ hoti hai, Freon nahi. [Q1337]
- ⚠ Trap: NTP pe molar volume 22.4 L/mol hota hai — mass/volume ratio proportion se molecular weight nikalte hain. [Q1349]